

ON THE UNIQUENESS OF THE EVEN PRIME: A SELF-CONTAINED DEVELOPMENT FROM THE PEANO AXIOMS

ABSTRACT. We give a self-contained proof that 2 is the unique even prime natural number. Working in informal classical set theory with a Dedekind–Peano structure $(\mathbb{N}, 0, S)$, we prove a recursion theorem (without presupposing the order relation), construct addition and multiplication, develop order and divisibility, and define primes, composites, and even numbers. After verifying that 2 itself is an even prime, we prove two complementary statements: no natural number strictly less than 2 is prime, and every even natural number strictly greater than 2 is composite. The second of these is proved in two genuinely different ways—by the divisor clause in the definition of primality, and by exhibiting a proper factorization. Combining these facts yields uniqueness. A companion LEAN 4 development checks the uniqueness layer on the built-in type `Nat`; we record precisely what is and is not machine-checked.

1. INTRODUCTION

Theorem 1.1 (Target). *If $p \in \mathbb{N}$ is prime and even, then $p = 2$. Equivalently: 2 is the unique even prime natural number.*

This classical fact is elementary, but a fully explicit account must say what a natural number is, construct arithmetic, define divisibility and primality, and justify every intermediate step. The present note supplies such an account, in the spirit of Landau’s *Foundations of Analysis* [1], while remaining short enough to be read as a single argument.

Two design choices distinguish the note. First, we take care with the ambient framework: induction is stated for arbitrary subsets of $\mathbb{N}$, and recursion is proved before order is defined, so that the construction of addition does not depend on a relation that itself depends on addition. Second, we carefully separate two *arithmetically distinct* proofs that every even $n > 2$ is composite from mere presentational variants of the same inference.

A companion formalization in LEAN 4 (stdlib only) checks the uniqueness theorem, the two primary proofs, definitional audits, and specification certificates on the kernel type `Nat`. It does *not* reconstruct $\mathbb{N}$ from Peano axioms inside the prover; see [Section 11](#).

2. AMBIENT FRAMEWORK AND THE PEANO AXIOMS

Convention. We work in informal classical set theory. In particular we allow formation of subsets of $\mathbb{N}$, functions $\mathbb{N} \rightarrow \mathbb{N}$, and finite sequences in $\mathbb{N}$. Induction will be stated for arbitrary subsets (Dedekind’s form), not as a first-order schema. Classical logic (including proof by contradiction) is used freely.

Definition 2.1 (Natural numbers). We postulate a set $\mathbb{N}$, a distinguished element $0 \in \mathbb{N}$, and a function $S: \mathbb{N} \rightarrow \mathbb{N}$ (the *successor*), subject to the axioms below. Elements of $\mathbb{N}$ are called *natural numbers*.

Axiom 2.2 (Zero is not a successor). For every $n \in \mathbb{N}$, $S(n) \neq 0$.

Axiom 2.3 (Injectivity of successor). For all $m, n \in \mathbb{N}$, if $S(m) = S(n)$, then $m = n$.

Date: July 30, 2026.

2020 Mathematics Subject Classification. Primary 11A41, 03B30; Secondary 00A35, 11A51.

Key words and phrases. Prime number, even integer, Peano axioms, divisibility, well-ordering, elementary number theory.

Axiom 2.4 (Induction). Let $A \subseteq \mathbb{N}$. Suppose $0 \in A$, and suppose that $n \in A$ implies $S(n) \in A$ for every $n \in \mathbb{N}$. Then $A = \mathbb{N}$.

Remark 2.5. When we say “let P be a property of natural numbers” we mean that $A = \{n \in \mathbb{N} : P(n)\}$ is a subset of $\mathbb{N}$ to which [Axiom 2.4](#) applies. This is the only comprehension principle we use without further comment.

Notation 2.6. Set $1 \stackrel{\text{def}}{=} S(0)$, $2 \stackrel{\text{def}}{=} S(1)$, $3 \stackrel{\text{def}}{=} S(2)$, and $4 \stackrel{\text{def}}{=} S(3)$.

Lemma 2.7. One has $0 \neq 1$ and $1 \neq 2$.

Proof. If $0 = 1 = S(0)$, then $S(0) = 0$, contradicting [Axiom 2.2](#). If $1 = 2$, then $S(0) = S(1)$, so $0 = 1$ by [Axiom 2.3](#). $\square$

Lemma 2.8 (Predecessor). For every $n \in \mathbb{N}$, either $n = 0$ or there is a unique $m \in \mathbb{N}$ with $n = S(m)$.

Proof. Let A be the set of n satisfying the claim. Then $0 \in A$. If $n \in A$, then $S(n) \in A$ by taking $m = n$. By [Axiom 2.4](#), $A = \mathbb{N}$. Uniqueness is [Axiom 2.3](#). $\square$

3. RECURSION, ADDITION, AND MULTIPLICATION

We prove recursion without mentioning the order relation $\leq$.

Lemma 3.1 (Recursion). Let $a \in \mathbb{N}$ and let $f : \mathbb{N} \rightarrow \mathbb{N}$. There exists a unique function $u : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$u(0) = a \quad \text{and} \quad u(S(n)) = f(u(n)) \text{ for all } n \in \mathbb{N}. \quad (3.1)$$

Proof. Uniqueness. If u and v both satisfy (3.1), the set $A = \{n : u(n) = v(n)\}$ contains 0 and is closed under S , so $A = \mathbb{N}$ by [Axiom 2.4](#).

Existence. We construct the graph of u as a set of pairs. Call a set $s \subseteq \mathbb{N} \times \mathbb{N}$ a *recursive sequence* for (a, f) if

- (i) $(0, a) \in s$, and
- (ii) whenever $(n, y) \in s$, also $(S(n), f(y)) \in s$.

Let $\mathcal{S}$ be the collection of all such s , and set

$$\Gamma \stackrel{\text{def}}{=} \bigcap_{s \in \mathcal{S}} s.$$

Then Γ itself is a recursive sequence. We claim: for every $n \in \mathbb{N}$ there is a unique $y \in \mathbb{N}$ with $(n, y) \in \Gamma$.

Existence of y : let A be the set of n for which some y has $(n, y) \in \Gamma$. Clearly $0 \in A$. If $n \in A$ with $(n, y) \in \Gamma$, then $(S(n), f(y)) \in \Gamma$, so $S(n) \in A$. Thus $A = \mathbb{N}$.

Uniqueness of y : let A be the set of n for which at most one y has $(n, y) \in \Gamma$. Then $0 \in A$: if $(0, y) \in \Gamma$ and $y \neq a$, the set $\Gamma \setminus \{(0, y)\}$ would still be a recursive sequence, contradicting minimality of Γ . If $n \in A$ and $(S(n), z_1), (S(n), z_2) \in \Gamma$, then there exist y_1, y_2 with $(n, y_i) \in \Gamma$ and $z_i = f(y_i)$; uniqueness at n forces $y_1 = y_2$, hence $z_1 = z_2$. So $S(n) \in A$, and $A = \mathbb{N}$.

Define $u(n)$ to be the unique y with $(n, y) \in \Gamma$. Then u satisfies (3.1). $\square$

Definition 3.2 (Addition). For each $m \in \mathbb{N}$, apply [Lemma 3.1](#) with $a = m$ and $f = S$ to obtain u_m . Set $m + n \stackrel{\text{def}}{=} u_m(n)$. Thus

$$m + 0 = m, \quad m + S(n) = S(m + n). \quad (3.2)$$

Definition 3.3 (Multiplication). For each $m \in \mathbb{N}$, apply [Lemma 3.1](#) with $a = 0$ and $f(x) = x + m$ to obtain v_m . Set $m \cdot n \stackrel{\text{def}}{=} v_m(n)$. Thus

$$m \cdot 0 = 0, \quad m \cdot S(n) = (m \cdot n) + m. \quad (3.3)$$

Lemma 3.4 (Arithmetic toolkit). *For all $a, b, c \in \mathbb{N}$:*

- (i) $0 + n = n$ and $n + 1 = S(n)$;
- (ii) $(a + b) + c = a + (b + c)$ and $a + b = b + a$;
- (iii) $0 \cdot n = 0$, $n \cdot 1 = n$, and $1 \cdot n = n$;
- (iv) $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ and $(a + b) \cdot c = (a \cdot c) + (b \cdot c)$;
- (v) $a \cdot b = b \cdot a$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

Proof. Each identity is proved by induction on the appropriate variable, using (3.2)–(3.3) and previously established clauses; cf. [1]*Ch. I. We record the inductive templates once: for a fixed identity $E(n)$, check $E(0)$ from the base clauses, then deduce $E(S(n))$ from $E(n)$ and the recursive step. Commutativity of addition uses the auxiliary fact $m + 1 = 1 + m$ (itself by induction), then inducts on the second summand. Distributivity and the remaining multiplicative laws follow similarly. $\square$

Lemma 3.5 (Additive cancellation). *If $x + c = y + c$, then $x = y$. Consequently, if $a + k = a$, then $k = 0$.*

Proof. Induct on c . The case $c = 0$ is immediate. If $x + S(c) = y + S(c)$, then $S(x + c) = S(y + c)$, so $x + c = y + c$ by Axiom 2.3, hence $x = y$ by the inductive hypothesis. For the consequence, take $y = 0$ and use $a + 0 = a$ with Lemma 3.4(i) after rewriting via commutativity: $a + k = a + 0 \Rightarrow k = 0$. $\square$

Lemma 3.6 (No zero-divisors). *If $a \neq 0$ and $a \cdot b = 0$, then $b = 0$. Equivalently: if $a \cdot b = 0$, then $a = 0$ or $b = 0$.*

Proof. Induct on b . If $b = 0$ we are done. If $b = S(b')$, then $a \cdot b = (a \cdot b') + a = 0$. Writing $a = S(a')$ by Lemma 2.8 (since $a \neq 0$), the sum $(a \cdot b') + S(a') = S((a \cdot b') + a')$ cannot be 0 by Axiom 2.2. $\square$

4. ORDER

Definition 4.1 (Order). Write $m \leq n$ if and only if $\exists k \in \mathbb{N}$ with $m + k = n$. Write $m < n$ if and only if $m \leq n$ and $m \neq n$. The relations $\geq$ and $>$ are the opposites.

Lemma 4.2 (Order toolkit). *For all $a, b, c \in \mathbb{N}$:*

- (i) $a \leq a$; if $a \leq b$ and $b \leq c$, then $a \leq c$;
- (ii) $a \leq b$ if and only if $a + c \leq b + c$;
- (iii) if $a \leq b$ and $b \leq a$, then $a = b$ (antisymmetry);
- (iv) $a < S(a)$; in particular $0 \leq a$;
- (v) $a < b$ if and only if $\exists k \neq 0$ with $a + k = b$;
- (vi) if $a \leq b$, then $S(a) \leq S(b)$;
- (vii) if $a < b$ and $b < c$, then $a < c$;
- (viii) if $a < b$, then $S(a) \leq b$ (discreteness).

Proof. (i) Take $k = 0$; for transitivity, add the witnesses and use Lemma 3.4(ii). (ii) The forward direction rearranges by associativity and commutativity. The converse uses Lemma 3.5. (iii) If $a + k = b$ and $b + \ell = a$, then $a + (k + \ell) = a$, so $k + \ell = 0$ by Lemma 3.5, hence $k = 0$ by Lemma 3.6 (or directly: $k \neq 0$ would make $k + \ell$ a successor). Thus $a = b$. (iv) Use $a + 1 = S(a)$ and $a \neq S(a)$ (proved by induction via Axioms 2.2 and 2.3). Then $0 \leq a$ with witness a . (v) Immediate from the definition and Lemma 3.5. (vi) If $a + k = b$, then $S(a) + k = S(b)$. (vii) Combine witnesses $k, \ell \neq 0$; then $k + \ell \neq 0$. (viii) Write $b = a + k$ with $k \neq 0$, so $k = S(k')$ and $b = S(a) + k'$. $\square$

Lemma 4.3 (Trichotomy). *For all $m, n \in \mathbb{N}$, exactly one of $m < n$, $m = n$, $m > n$ holds.*

Proof. Existence. Fix m and induct on n . For $n = 0$: if $m = 0$ we are done; if $m \neq 0$ then $0 < m$ by Lemma 4.2(iv)–(v). Assuming the claim for n , the three cases for (m, n) upgrade to cases for $(m, S(n))$ by Lemma 4.2(iv),(viii).

Exclusivity. The pairs involving equality contradict $m \neq n$ in the definition of $<$. If $m < n$ and $n < m$, write $m + k = n$ and $n + \ell = m$ with $k, \ell \neq 0$; then $m + (k + \ell) = m$, so $k + \ell = 0$ by Lemma 3.5, impossible since $k \neq 0$. $\square$

Lemma 4.4. *One has $n < 2$ if and only if $n = 0$ or $n = 1$. Consequently, $n > 1$ if and only if $n \geq 2$.*

Proof. If $n = 0$ or $n = 1$, then $n < 2$ by Lemma 4.2(iv),(vii). Conversely, if $n < 2$ and $n \neq 0$, then $n = S(m)$ and $m + 1 < 1 + 1$, so $m < 1$ by Lemma 4.2(ii) for $<$. The only natural number < 1 is 0, so $n = 1$.

For the second claim: if $n \geq 2$ then $n > 1$. If $n > 1$, then $n \neq 0, 1$, so $\neg(n < 2)$ by the first claim, hence $n \geq 2$ by trichotomy. $\square$

Theorem 4.5 (Well-ordering). *Every nonempty $S \subseteq \mathbb{N}$ has a least element.*

Proof. Let $P(n)$ assert that every subset of $\mathbb{N}$ meeting $\{k : k \leq n\}$ has a least element. Then $P(0)$ holds because the only candidate is 0, which is least in $\mathbb{N}$ by Lemma 4.2(iv). Assuming $P(n)$, let S meet $\{k : k \leq S(n)\}$. If S meets $\{k : k \leq n\}$, apply $P(n)$. Otherwise every element of S exceeds n , yet some $s \in S$ has $s \leq S(n)$. By Lemma 4.2(viii) and antisymmetry, $s = S(n)$, which is then least in S . Induction gives $P(n)$ for all n ; for nonempty S pick $s \in S$ and apply $P(s)$. $\square$

5. DIVISIBILITY

Definition 5.1. Write $d \mid n$ if $\exists k \in \mathbb{N}$ with $n = d \cdot k$.

Lemma 5.2 (Divisibility toolkit). (i) $a \mid a$, $1 \mid a$, and $a \mid 0$;
(ii) if $a \mid b$ and $b \mid c$, then $a \mid c$;
(iii) if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
(iv) if $d \mid n$ and $n \neq 0$, then $d \neq 0$ and $d \leq n$;
(v) if $n = d \cdot k$ with $d > 1$ and $k > 1$, then $1 < d < n$ and $1 < k < n$.

Proof. (i)–(iii) are immediate from Lemma 3.4. (iv) If $d = 0$ then $n = 0$ by Lemma 3.4(iii). Writing $n = d \cdot S(k') = (d \cdot k') + d$ gives $d \leq n$. (v) Apply (iv) and Lemma 3.6: neither factor is 0 or 1, and $d = n$ would force $k = 1$. $\square$

6. PRIMES, COMPOSITES, AND EVEN NUMBERS

Definition 6.1 (Prime). A number p is *prime* if $p > 1$ and every divisor of p is 1 or p .

Definition 6.2 (Composite). A number n is *composite* if $n > 1$ and n is not prime.

Definition 6.3 (Even). A number n is *even* if $2 \mid n$.

Remark 6.4 (Prime versus irreducible). Definition 6.1 is the *irreducible* condition in the monoid $(\mathbb{N}, \cdot)$: after noting that $uv = 1$ forces $u = v = 1$ (so the only unit is 1), it says p has no nontrivial factorization. We never use the Euclid property $p \mid ab \Rightarrow p \mid a \vee p \mid b$.

Lemma 6.5. *A number $n > 1$ is composite if and only if there exist $d, k \in \mathbb{N}$ with $n = d \cdot k$, $1 < d < n$, and $1 < k < n$.*

Proof. If such d, k exist, then $d \mid n$ with $d \neq 1$ and $d \neq n$, so n is not prime, hence composite. Conversely, if n is composite then some divisor d satisfies $d \neq 1$ and $d \neq n$. By Lemma 5.2(iv) and Lemma 4.4, one has $1 < d < n$. Writing $n = d \cdot k$ and applying Lemma 5.2(v) gives $1 < k < n$. $\square$

Example 6.6. (a) 0 is even, since $0 = 2 \cdot 0$, but not prime ($0 \not> 1$).

(b) 1 is neither even nor prime.

(c) 2 will be shown to be an even prime (Proposition 6.8).

- (d) 3 is prime: its only divisors are 1 and 3 (any divisor d satisfies $d \leq 3$ and $d \neq 0$, hence $d \in \{1, 2, 3\}$, and $2 \nmid 3$).
- (e) 4 is composite: $4 = 2 \cdot 2$ with $1 < 2 < 4$.

Lemma 6.7. *If $d \mid 2$, then $d = 1$ or $d = 2$.*

Proof. By Lemma 5.2(iv), $d \neq 0$ and $d \leq 2$. By Lemma 4.4 and trichotomy, $d = 1$ or $d = 2$. $\square$

Proposition 6.8 (Two is an even prime). *The number 2 is prime and even.*

Proof. Evenness: $2 = 2 \cdot 1$. Primality: $2 > 1$, and every divisor is 1 or 2 by Lemma 6.7. $\square$

Lemma 6.9 (Halving). *If $n > 2$ and $n = 2 \cdot k$, then $1 < k < n$.*

Proof. If $k = 0$ then $n = 0$, contradicting $n > 2$. If $k = 1$ then $n = 2$, again a contradiction. Thus $k > 1$. Also $2 > 1$, so Lemma 5.2(v) gives $k < n$. $\square$

7. NO PRIME LESS THAN TWO

Theorem 7.1 (No prime below two). *If $n < 2$, then n is not prime. A fortiori, no even prime is strictly less than 2.*

Proof. By Lemma 4.4, $n = 0$ or $n = 1$; neither satisfies $n > 1$. $\square$

8. NO EVEN PRIME GREATER THAN TWO

Theorem 8.1 (Even numbers above two are composite). *If $n > 2$ and n is even, then n is not prime. Equivalently, every even natural number strictly greater than 2 is composite.*

We give two proofs. They rest on different arithmetic ideas: the first uses only the divisor clause in Definition 6.1; the second exhibits an explicit proper factor.

8.1. Proof I: definitional.

Proof I of Theorem 8.1. We first prove the equivalent uniqueness form

$$\forall p \in \mathbb{N}, \quad \text{Prime}(p) \wedge \text{Even}(p) \Rightarrow p = 2,$$

and then deduce the theorem.

Let p be prime and even. Then $2 \mid p$. By Definition 6.1, $2 = 1$ or $2 = p$. The first alternative is false by Lemma 2.7, so $p = 2$.

Now if $n > 2$ is even and (toward a contradiction) prime, the claim just proved forces $n = 2$, contradicting $n > 2$. Since $n > 2 > 1$, the number n is composite. $\square$

Remark 8.2 (Presentation variants). The same divisor-clause inference may be packaged as a contradiction (“assume $p \neq 2$ ”) or routed through the observation that if primes p, q satisfy $p \mid q$ then $p = q$. These are not independent proofs of Theorem 8.1; they rearrange Proof I.

8.2. Proof II: factorization.

Proof II of Theorem 8.1. Let $n > 2$ be even, so $n = 2 \cdot k$ for some k . By Lemma 6.9, $1 < k < n$. Thus $k \mid n$ with $k \neq 1$ and $k \neq n$, so n fails Definition 6.1. Since $n > 1$, n is composite by Definition 6.2 (or Lemma 6.5). $\square$

Remark 8.3 (Least counterexamples are idle here). Theorem 4.5 would supply a least even prime > 2 if any existed, but the factorization in Proof II already rules out *every* such candidate. Minimality is never used, so we do not record a separate least-counterexample proof.

Corollary 8.4. *If p is prime and even, then $p = 2$.*

Proof. Contained in Proof I; alternatively, combine Theorem 8.1 with $p > 1 \Rightarrow p \geq 2$ from Lemma 4.4 and trichotomy. $\square$

Lemma 8.5 (Equivalence of formulations). *The following are equivalent:*

- (i) $\forall n \in \mathbb{N}$, if $n > 2$ and n is even, then n is not prime;
- (ii) $\forall p \in \mathbb{N}$, if p is prime and even, then $p = 2$.

Neither statement by itself asserts that 2 is prime or even.

Proof. (i) $\Rightarrow$ (ii): if p is an even prime and $p > 2$, contradict (i); since $p > 1$, trichotomy yields $p = 2$.
(ii) $\Rightarrow$ (i): if $n > 2$ is even and prime, then (ii) forces $n = 2$. $\square$

9. THE UNIQUENESS THEOREM

Theorem 9.1 (Uniqueness of the even prime). *The following hold, and moreover (A) is equivalent to (B):*

- (A) 2 is prime and even, and every even prime equals 2;
- (B) there is no even prime $n < 2$, there is no even prime $n > 2$, and 2 is an even prime;
- (C) every even prime equals 2 (uniqueness only);
- (D) there is no even prime other than 2 (uniqueness only, negative form).

Statements (C) and (D) are equivalent to each other and to Lemma 8.5, but they do not imply (A) without the separate witness that 2 is an even prime.

Proof. Proposition 6.8 gives that 2 is an even prime. Theorem 7.1 gives that no prime (hence no even prime) is < 2 . Theorem 8.1 gives that no even prime is > 2 . Corollary 8.4 gives that every even prime equals 2. This establishes (A) and (B), which are equivalent by trichotomy on any even prime.

The equivalence of (C) and (D) is pure quantifier logic, and their equivalence to Lemma 8.5 is that lemma. Finally, (C) alone does not yield that 2 is prime: one needs Proposition 6.8 (or an equivalent witness) to upgrade (C) to (A). $\square$

Corollary 9.2. *The unique even prime natural number is 2.*

10. DEPENDENCE

The external assumptions are: informal classical set theory; a Dedekind–Peano structure $(\mathbb{N}, 0, S)$. No external theorem of number theory is used.

The logical spine is

Peano $\rightarrow$ recursion $\rightarrow (+, \cdot) \rightarrow$ order $\rightarrow$ divisibility $\rightarrow$ primes $\rightarrow$ Theorems 7.1, 8.1 and 9.1.

Well-ordering is developed for completeness but is not required for the uniqueness proof (see Remark 8.3). Of the two proofs of Theorem 8.1, Proof I uses only the divisor clause and $2 \neq 1$; Proof II uses the halving lemma and the definition of composite.

11. COMPANION LEAN FORMALIZATION

A companion development in LEAN 4 (stdlib only, no Mathlib) checks the uniqueness layer on the kernel type `Nat`. In particular it establishes:

- Prime, Composite, Even, and `nd Irreducible`, with Prime $\leftrightarrow$ Irreducible and the proper-factorization characterization of composites;
- the witness that 2 is an even prime, together with the examples 0 even, 1 not even, 3 prime, 4 composite;
- Proofs I and II of Theorem 8.1, and the repaired equivalences of Theorem 9.1;
- a specification certificate separating uniqueness-only claims from existence+uniqueness.

The Lean code does *not* reconstruct $\mathbb{N}$ from the Peano axioms; it takes `Nat` as given and uses kernel automation (e.g. `omega`) for routine inequalities. Thus the machine-checked artifact certifies the uniqueness argument relative to Lean’s $\mathbb{N}$, while Sections 2 to 4 supply the narrative Peano development.

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